

Analysis of Scientific Data



Lecture 3

Exploring Data

Course Reading List:

	The R Project for Statistical Computing Website - Required ▾	VIEW ONLINE	↕	⋮
ADD RESOURCE / ADD PARAGRAPH / ADD SECTION				
	RStudio Website - Required ▾	VIEW ONLINE	↕	⋮
	A Portable Introduction to Data Analysis Book - by Michael Bulmer - 2024 - Recommended ▾ Free open textbook UQ author	VIEW ONLINE	↕	⋮
	Chapter 18: Core issues in ethics for scientists in Applied Ethics: Strengthening Ethical Practices Chapter - by Julian Lamont; Kevin Lowe - 2012 - Required ▾	VIEW ONLINE	↕	⋮
	Mind on statistics: Australian & New Zealand Book - by Helen MacGillivray; Jessica M. Utts; R. F. Heckard - 2014 - Second Australian and New Zealand edition - Recommended ▾	VIEW ONLINE	↕	⋮
	Chance encounters: a first course in data analysis and inference Book - by Christopher John Wild; George Arthur Frederick Seber - 2000 - Further ▾	VIEW AVAILABILITY	↕	⋮

learnR tutorial: <https://learnr.numbat.space/>

Recap

- types of variables
- types of studies
- null hypothesis
- p -value

Heart failure clinical records

In this lecture we will explore data from the medical records of 299 patients who had heart failure, collected during their follow-up period.

Chicco, D. & Jurman, G. (2020) Machine learning can predict survival of patients with heart failure from    alone.

BMC Medical Informatics and Decision Making **20**, 16.

Heart failure clinical records

age	age of the patient (years)
anaemia	decrease of red blood cells
hpb	patient has hypertension
cpk	level of the creatinine phosphokinase enzyme in the blood ($\mu\text{g/L}$)
diabetes	patient has diabetes
ejection	fraction of blood leaving the heart at each contraction (%)
platelets	platelets in the blood (kiloplatelets/mL)
sex	M/F
creatinine	level of serum creatinine in the blood (mg/dL)
sodium	level of serum sodium in the blood (mEq/L)
smoking	patient smokes
time	follow-up period (days)
death	patient deceased during the follow-up period)

Poll Question

Is this study observational or experimental?

1. Observational
2. Experimental

Poll Question

What type of variable is **age**?

1. Continuous
2. Discrete
3. Nominal
4. Ordinal

Poll Question

What type of variable is **creatinine**?

1. Continuous
2. Discrete
3. Nominal
4. Ordinal

Visualising Distributions

The pattern of variability we see is called the *distribution* of the variable.

For quantitative variables we want to describe

- Location (or central tendency)
- Spread
- Shape
- Deviations from the overall pattern

Take notes about what tools are useful in different situations to visualise and summarise different variables.

base

```
hist(heart$creatinine)
```

lattice

```
histogram(~creatinine, heart)
```

ggplot2

```
ggplot(heart, aes(creatinine)) + geom_histogram()
```

Poll Question

How would you describe the distribution of **creatinine**?

1. Symmetric
2. Skewed to the right
3. Skewed to the left

Poll Question

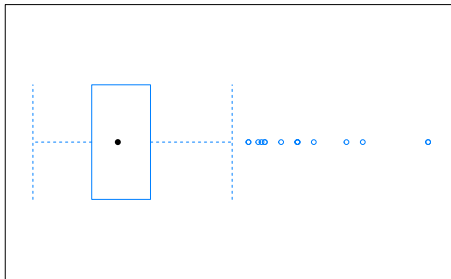
What is the interquartile range for **ejection**?

1. 10
2. 15
3. 20
4. 25

Poll Question

How would you describe the distribution of the variable shown in this box plot?

1. Symmetric
2. Skewed to the right
3. Skewed to the left



For two quantitative variables, we describe the relationship seen in a scatter plot in terms of

- Direction – is the trend positive or negative?
- Linearity – is the trend a line or a curve?
- Strength – how much variability is there relative to the trend?

Numerical Measures

Measures of location:

- mean
- mode
- median

Measures of variability:

- range
- IQR
- variance
- standard deviation